# JC3504 Robot Technology

Lecture 10: [Extension] Biology and Robotics

# Outline

- Bio-inspired Robotics
- Biological Materials for Robotics
- Human–Robot Interaction (Human-in-the-loop)

# An Alternative View

In biological systems, intelligence does not reside solely in the “brain”.

It is often distributed across the body, the materials, and the surrounding environment.

Structure, physical dynamics, and interaction with the world all contribute to adaptive behaviour.

# Bio-inspired Robotics

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# Bio-inspired Robotics

Bio-inspiration is not about copying animals, but about understanding why certain designs are effective.

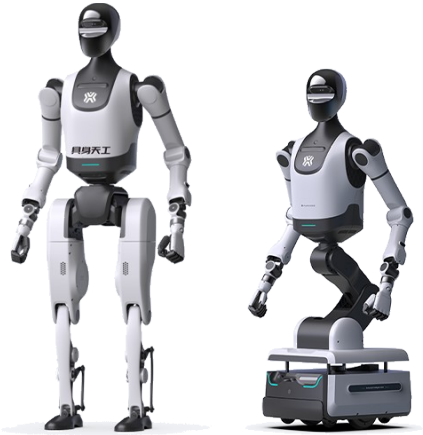
In engineering, this means exploiting structure and physical dynamics to reduce the need for complex computation.

When environments are unpredictable, simple and rapid responses can be more reliable than precise global planning.

- Morphology-inspired Robotics
- Behaviour-inspired Robotics

# Morphology-inspired Robotics

The most direct form of bio-inspired robotics is to replicate the physical structures found in animals.
By adopting similar body configurations, robots can exploit proven solutions for locomotion, balance and environmental interaction.



Humanoid robots

Quadruped robots

Snake robots

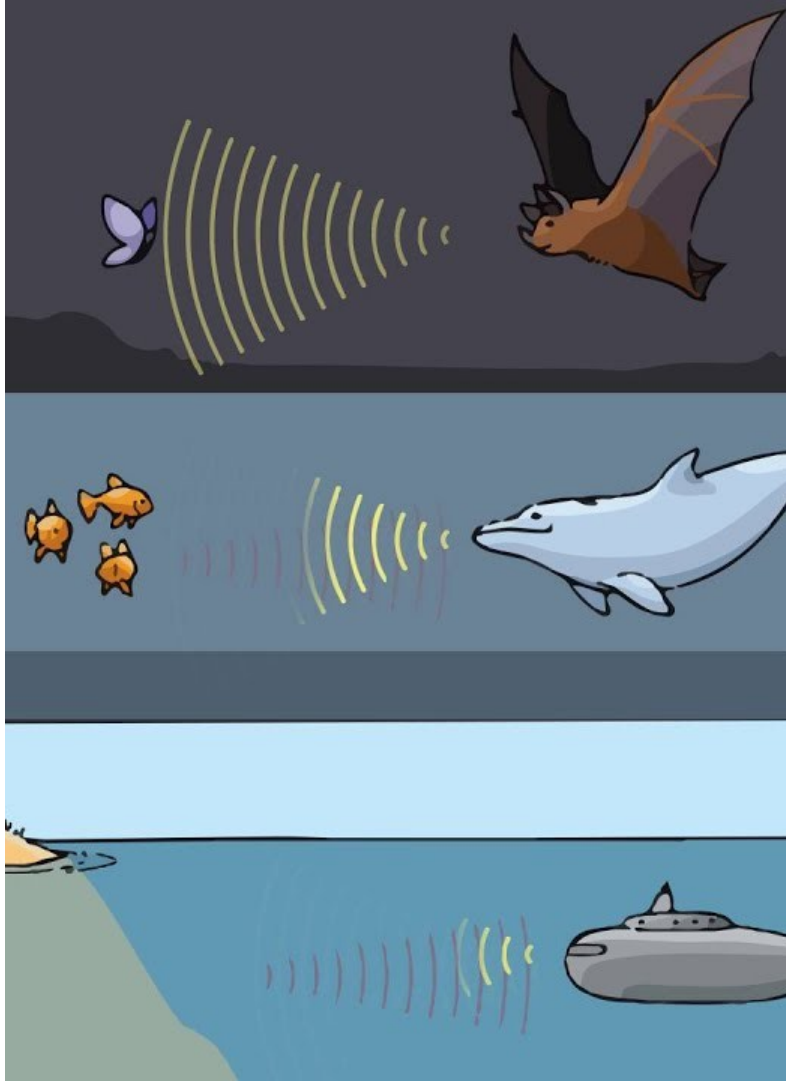
# Morphology-inspired Robotics

The real insight lies in understanding the structural intelligence that biological systems have acquired through evolution.

These forms are not arbitrary; they embed mechanical efficiency, energy optimisation and adaptive behaviour within their physical design.

# Ultrasonic Sensors *(review)*

- These measure distance by emitting ultrasonic pulses and receiving the reflected waves.
- The principle is based on the time difference between the emission and reception of the ultrasonic pulses, calculating the distance using the speed of sound. Ultrasonic sensors work well in air and are commonly used for indoor robot obstacle avoidance and distance measurement.



# Stickybot

Developed at Stanford University, Stickybot was inspired by the millions of microscopic hairs (setae) found on a gecko's feet.
These nano-scale structures generate dry adhesion through van der Waals forces, enabling strong yet reversible attachment.

Stickybot can climb smooth vertical surfaces without glue or suction cups, and it leaves no residue behind. The adhesive capability arises from surface structure rather than chemical bonding.

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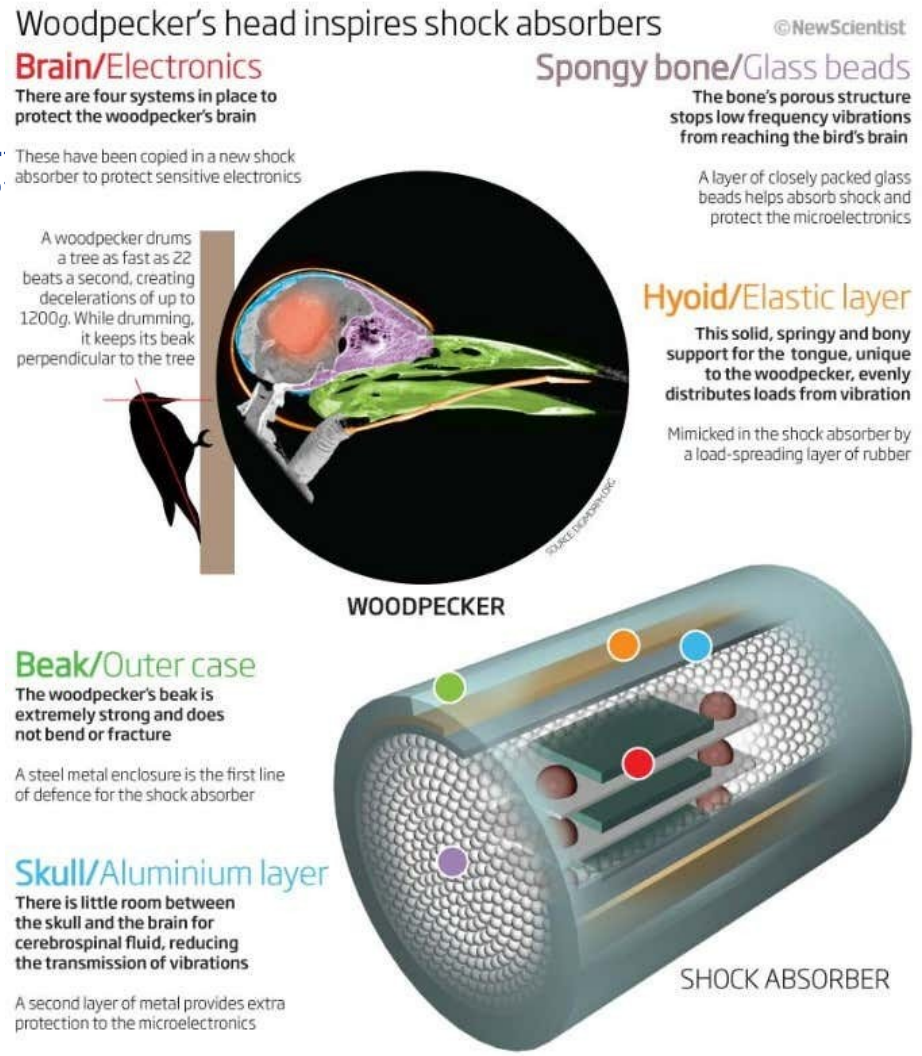
# Woodpecker-inspired Desig

Researchers at Harbin Institute of Technology studied how woodpeckers endure repeated high-speed impacts without neurological damage (around 20 Hz and up to 1200 g ).

They found that the bird's skull incorporates a hierarchical, porous structure combining rigidity and compliance.

This morphology effectively dissipates impact energy, illustrating how structural design can provide passive shock absorption without active control.

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# Behaviour-inspired Robotics

The core idea of behaviour-inspired robotics is that a system need not fully understand the world before it can act.

Instead, it can act directly and adapt through continuous interaction with its environment.

When environments are unpredictable, simple and rapid responses are often more reliable than precise global planning.

# Caenorhabditis elegans

Approximately 1 mm in length; Contains only 302 neurons in total.

Studies show that during chemotaxis (e.g. moving towards a food odour), its core decision-making mechanism relies primarily on just two sensory neurons (such as AWC) to detect changes in odour concentration.

- If the odour becomes stronger → reduce turning → continue forward
- If the odour becomes weaker → increase turning → change direction

In other words, the worm does not construct a map of its environment. It navigates through instantaneous gradient detection combined with a simple feedback loop.

# German cockroach

Escape response latency: approximately 40 ms. The cockroach does not “analyse” the type of threat. Instead:

- Its cerci detect sudden air movement
- A dedicated neural circuit is immediately triggered
- It accelerates away from the direction of stimulation
- There is no modelling, no path planning, and no internal environmental representation.

Its effectiveness comes from:

- Extremely fast local sensing
- Direct coupling between perception and motor output
- No requirement for global understanding

# Red imported fire ants

Fire ants form bridges and rafts (classic examples of emergent behaviour), no single ant understands:

- The global structure
- The final shape
- The overall system objective

Each ant simply follows local rules:

- Encounter a gap → grip
- Experience pressure → stabilise
- Information propagates through local interactions

Yet stable, load-bearing structures emerge from these simple behaviours.

# Genghis (1989) by Rodney Brooks, MIT

No global map; No central planner

Instead, the system consisted of multiple simple behavioural modules layered together:

- Obstacle avoidance
- Forward locomotion
- Leg lifting
- Terrain probing

The robot did not “think” in a deliberative sense. Its behaviour emerged from layered reflex networks operating in parallel.

# Genghis (1989) by Rodney Brooks, MIT

< Video: 10 - Genghis-II.mp4>

# Genghis (1989) by Rodney Brooks, MIT

This example suggests that locomotion can be treated as a physical phenomenon rather than purely as a control problem.

The interaction between body and environment performs part of the "computation" required for movement.

# Biological Materials for Robotics

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10.1109/IEMBS.2009.5334433

# Manduca sexta: A Bio-hybrid Flight System

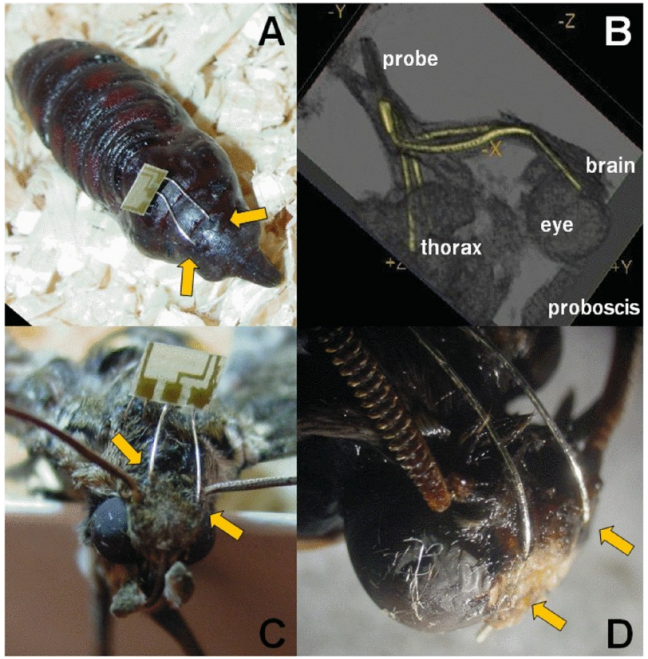
In this approach, the insect itself functions as a biological actuation and flight platform.

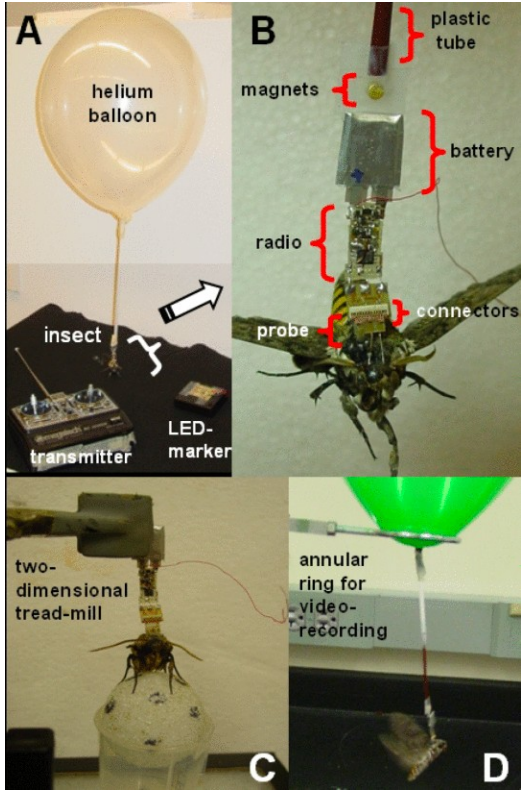
Rather than constructing artificial wings and actuators, researchers interface with the insect's existing neuromuscular system through implanted micro-electrodes.

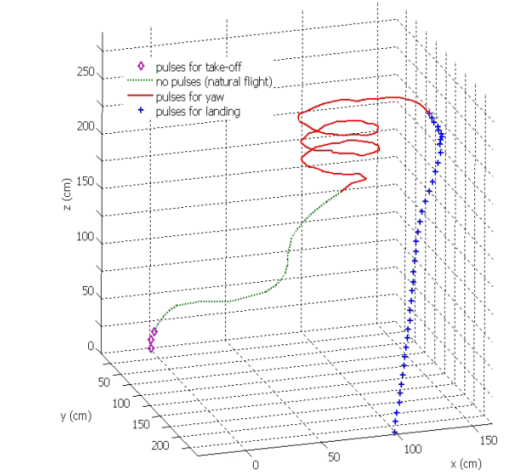
Wireless stimulation enables partial control over flight behaviours such as take-off, directional yaw, and landing.

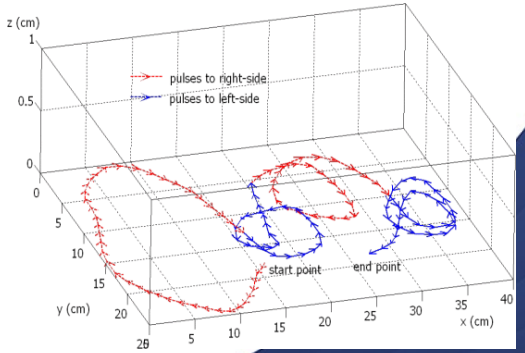
This represents a bio-hybrid system in which biological and engineered components form a single functional unit.

# Manduca sexta: A Bio-hybrid Flight System

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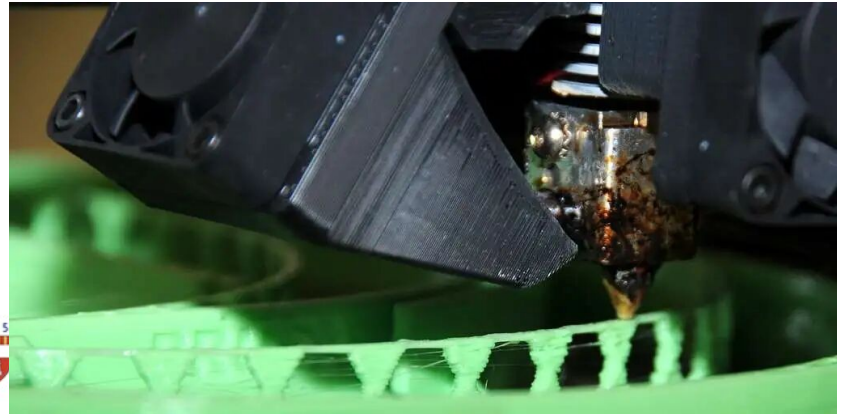
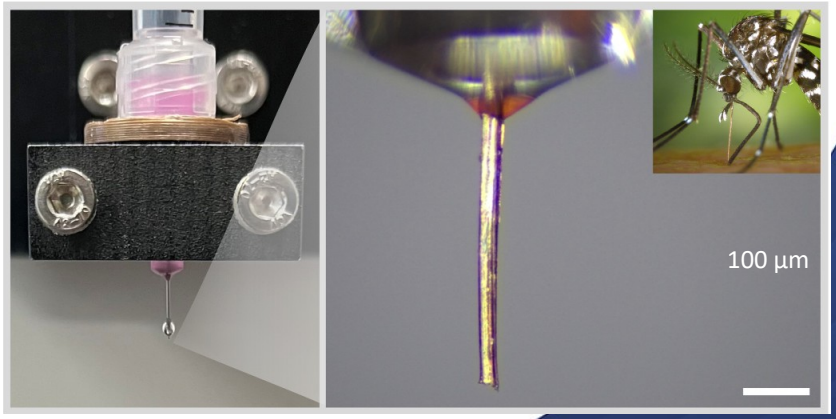
10.1126/sciadv.adw9953

# Mosquito Proboscis as a 3D Printing Nozzle

Direct Ink Writing (DIW) for high-resolution printing and bioprinting requires ultra-fine micro-nozzles.

The mosquito proboscis, with a diameter of approximately 20 µm, features a smooth, straight internal channel that allows fluid to flow with minimal clogging or turbulence. Composed primarily of chitin, this biological structure combines high strength with flexibility, enabling it to withstand pressures of up to around 60 kPa during fluid extrusion.





# Why Is MIT Making Robot Insects

< Video: 10 - Mosquito Proboscis Dispensing Tip.mp4>

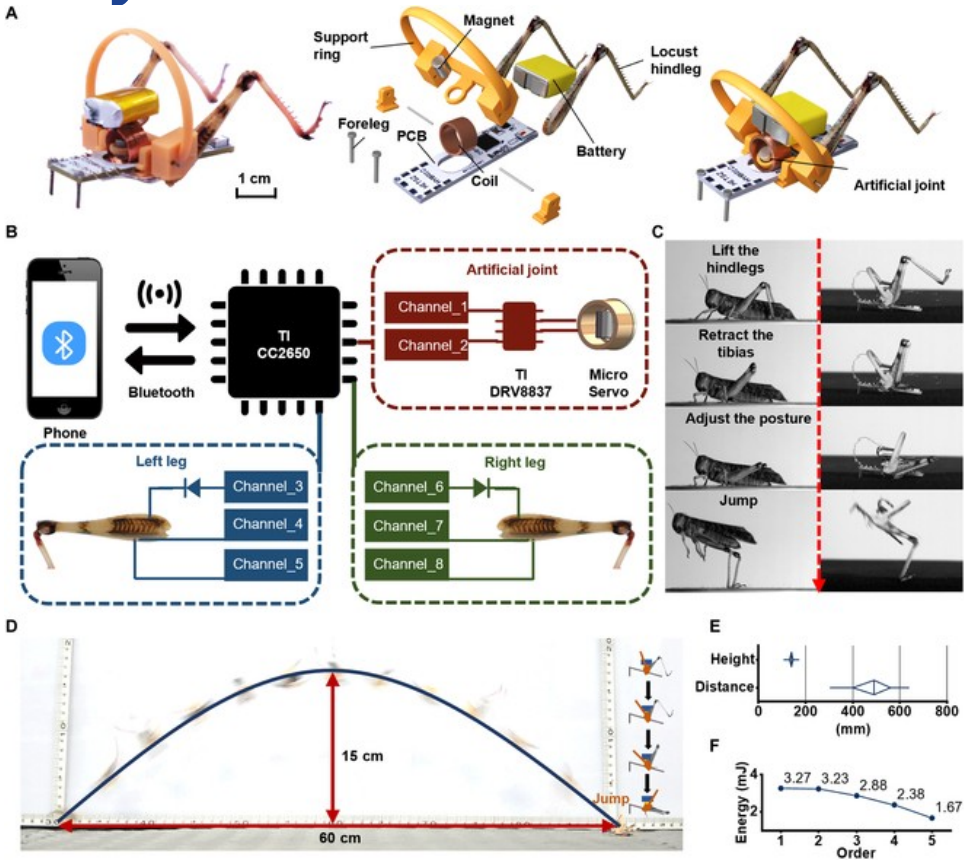
10.34133/research.0943

# Locust Hindleg as a Bio-hybrid Actuator

Electrical stimulation of isolated locust hind legs activates the natural flexor–extensor muscle system, enabling high-power, rapid jumping.

Integrated with artificial structural components and electronic control, the system demonstrates how biological muscle can serve as an ultra-efficient actuation mechanism.

This illustrates that biological tissue can provide high power-to-weight ratios that are difficult to achieve with conventional micro-actuators.

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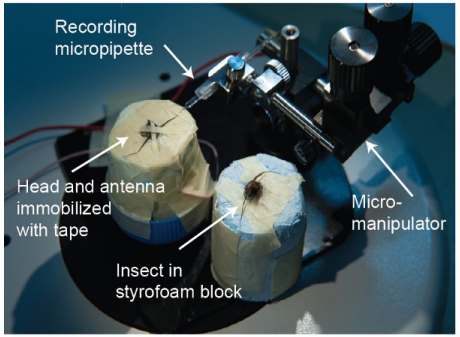
10.3791/51704

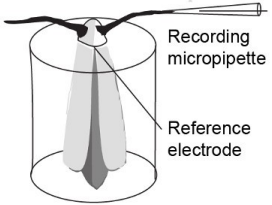
# Insect Antennae as Bio-derived Chemical Sensors

**Application:** A high-bandwidth olfactory search platform for chemical leak detection and odour source localisation.

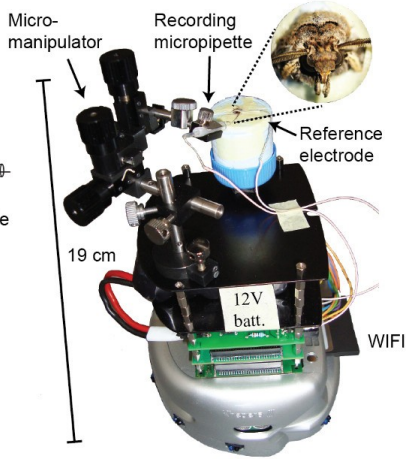
The system is capable of resolving intermittent odour pulses of up to around 10 Hz, exceeding the recovery speed of many conventional artificial gas sensors.

**A** EAG preparation




**B** Robotic setup

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# Why Is MIT Making Robot Insects

< Video: 10 - Why Is MIT Making Robot Insects.mp4>

# Advantages and Limitations

Advantages:

- Extremely high power-to-weight ratio (e.g. biological muscle actuation)
- Highly efficient micro-structures refined by evolution
- Superior sensing capabilities in certain domains (e.g. olfaction)

Limitations:

- Biological materials cannot self-repair once removed from the organism
- Many components are prone to ageing, degradation, and decay
- Limited lifespan and reproducibility
- Difficulties in standardisation and large-scale deployment

# Human–Robot Interaction (Human-in-the-loop)

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# Human-in-the-loop

Human-in-the-loop systems treat the human not as an external operator, but as part of the control architecture.

Intelligence emerges from the closed interaction between human cognition and machine execution.

In many real-world domains, full autonomy is less constrained by algorithmic capability than by safety, responsibility, robustness to rare events, and practical deployment considerations.

A robotic system = machine capability + human judgement

# Human-in-the-loop

A typical human-in-the-loop system can be represented as:

Human (intention / strategy)
→ Input interface (joystick, pedals, eye tracking, EMG, speech)
→ Robot control and autonomy modules (prediction, constraint handling, planning, control)
→ Environment / task outcome
→ Multimodal feedback (visual, haptic, status indicators)
→ Human updates strategy

# Da Vinci Surgical System

Surgeon's hands
→ Master console manipulators→ Master–slave mapping (motion scaling and tremor filtering)→ Robotic surgical arms→ High-definition 3D endoscopic visual feedback→ Surgeon adjusts movements

**Advantages:**

- Enables remote and minimally invasive surgery
- Delegates high-precision execution to the robotic system

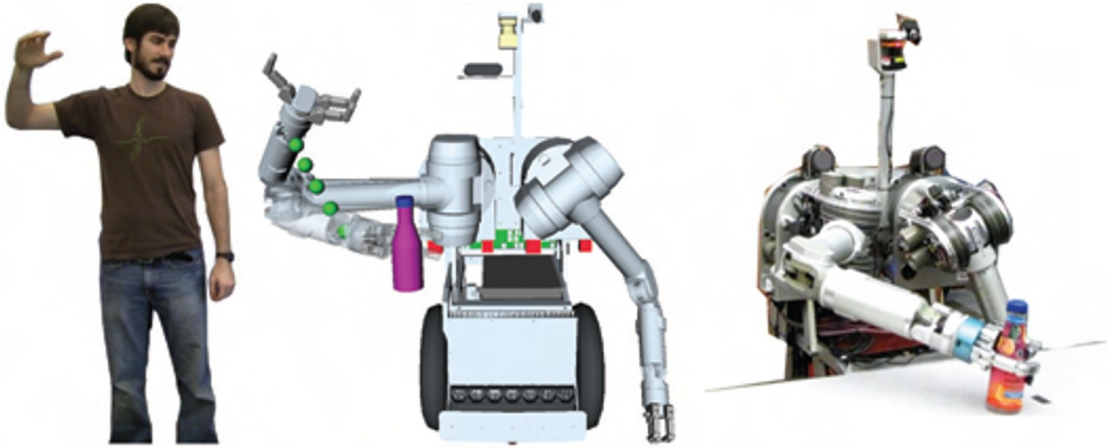
**Limitations:** Limited haptic (tactile) feedback

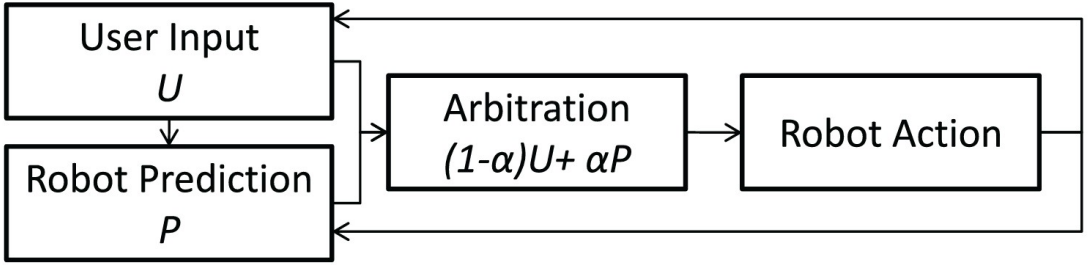
10.1177/0278364913490324

# Policy Blending in Shared Control

Users operate a robotic arm through an imperfect interface tc perform grasping and manipulation tasks.
The robot predicts user intent an selectively intervenes to assist.

The human provides directional control and real-time corrections while the robot handles intent prediction and supportive motior adjustment.



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# Human augmentation

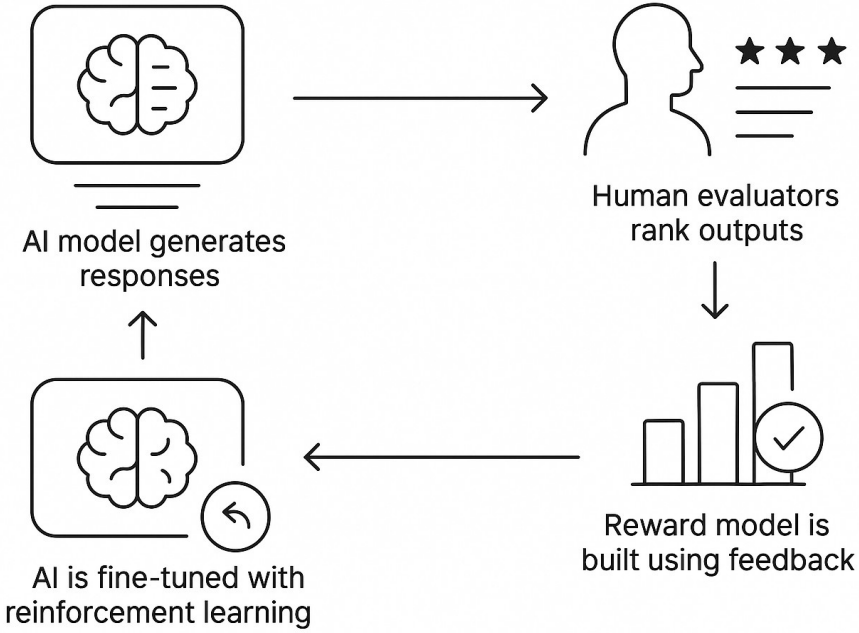
Human augmentation refers to systems that enhance human physical or cognitive abilities through robotic or computational support.

Instead of replacing humans, these systems integrate humans into the control loop to amplify strength, perception, or decision-making.

# Reinforcement Learning from Human Feedback

Reinforcement Learning from Human Feedback (RLHF) is a learning framework where human preferences are used to guide policy optimisation.

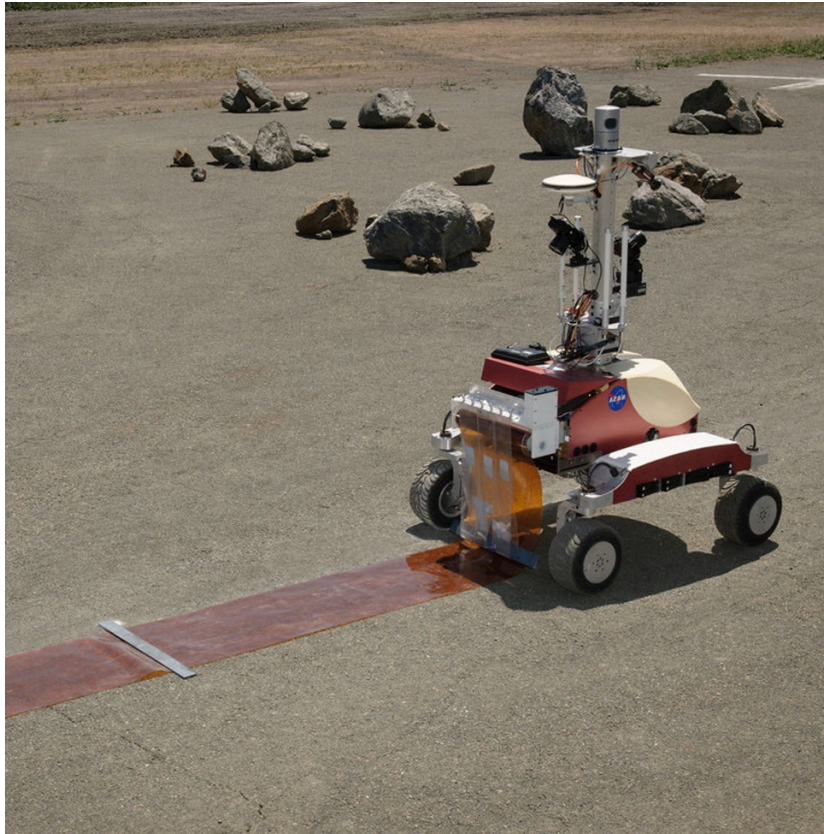
Instead of relying solely on predefined reward functions, the system learns what is desirable from human evaluations.



# Supervised Autonomy under Communication Delay

In second-scale communication delay scenarios (e.g. teleoperation of remote robots or planetary rovers), the system adopts a model of supervised autonomy with intermittent teleoperation.

The operator issues task sequences and high-level goals. The robot executes autonomously with interactive supervision



# Advantages and Limitations

## Advantages

- Combines human judgement with machine precision
- Improved robustness to rare or unexpected events
- Clear accountability in safety-critical applications
- Faster deployment compared to fully autonomous systems

## Limitations

- Human attention and cognitive load constraints
- Slower response compared to fully autonomous control in time-critical scenarios
- Interface design complexity

# Conclusion

Robotics is not only about algorithms and control. Intelligence can emerge from:

- Structure: morphology inspired by biological design
- Dynamics: body–environment interaction
- Local rules: behaviour-based control
- Biological integration: leveraging natural materials and systems
- Human integration: embedding human judgement within control loops

Across these perspectives, a common theme appears:

Intelligence is often distributed, across body, material, environment, and human–machine interaction, rather than confined to a central controller.

Effective robotic systems are therefore not defined solely by autonomy, but by how computation, physics, biology, and human cognition are combined.